Module 1

Q1.(Practical 1 & 2 based Q)
Q1. Read a Text File with Header (4 Marks)
Write an R program to read a text file with header and display its contents. Q2. Import CSV File (4 Marks)
Write an R program to import a CSV file and display it.
Q3. Read Excel File using readxl Package (5 Marks)
Write R code to:
● Install and load readxl package
● Read an Excel file
● Display the imported data
—---------- Q1. Create Matrix (4 Marks)
Write an R program to create a matrix with numeric values.
Q2. Display Dimensions (4 Marks)
Write an R program to display the dimensions of the matrix using dim().
Q3. Summary and Head (5 Marks)
Write an R program to:
Create a dataset
Display summary using summary()
Display first rows using head()

Ans :-

```
# -------------------------------
# Practical 1 : File Handling
# ------------------------------

# Q1. Read a Text File with Header
text_data <- read.table("data.txt", header = TRUE)
print(text_data)

# Output:
#   ID   Name Marks
# 1  1 Sujal    78
# 2  2 Amit     85
# 3  3 Riya     90


# Q2. Import CSV File
csv_data <- read.csv("data.csv")
print(csv_data)

# Output:
#   ID   Name Marks
# 1  1 Sujal    78
```

```
# 2  2 Amit     85
# 3  3 Riya     90


# Q3. Read Excel File using readxl
install.packages("readxl")
library(readxl)

excel_data <- read_excel("data.xlsx")
print(excel_data)

# Output:
#   ID   Name Marks
# 1  1 Sujal    78
# 2  2 Amit     85
# 3  3 Riya     90


# -------------------------------
# Practical 2 : Matrix + Dataset
# -------------------------------

# Q1. Create Matrix
mat <- matrix(c(10, 20, 30, 40, 50, 60), nrow = 2, ncol = 3)
print(mat)

# Output:
#      [,1] [,2] [,3]
# [1,]   10   30   50
# [2,]   20   40   60


# Q2. Display Dimensions
dim(mat)
# [1] 2 3


# Q3. Summary and Head
student_data <- data.frame(
  ID = c(1, 2, 3, 4, 5),
  Name = c("Sujal", "Amit", "Riya", "Neha", "Raj"),
  Marks = c(78, 85, 90, 88, 76)
)

summary(student_data)
#        ID        Name               Marks
# Min.   :1   Length:5           Min.   :76.0
# 1st Qu.:2   Class :character   1st Qu.:78.0
```

```
# Median :3   Mode  :character   Median :85.0
# Mean   :3                      Mean   :83.4
# 3rd Qu.:4                      3rd Qu.:88.0
# Max.   :5                      Max.   :90.0

head(student_data)
#   ID  Name Marks
# 1  1 Sujal    78
# 2  2  Amit    85
# 3  3  Riya    90
# 4  4  Neha    88
# 5  5   Raj    76
```

Q2.(Practical 3 based q)
Q1. Install and Load dplyr (5 Marks)
Write R code to:
● Install and load dplyr package
● Display dataset using glimpse()
Q2. Unique Values and Frequency Table (4 Marks)
Write an R program to:
● Display unique values from Name column
● Create frequency table of Age
Q3. Tibble Conversion (4 Marks)
Write R code to:
● Install and load tibble package
● Convert dataset into tibble
● Display the tibble dataset
—---------- Q1. Create Employee Dataset (5 Marks)
Create a dataset in R with fields:
● Employee_ID
● Employee_Name
● Department
● Salary
Display the dataset using print() and View().
Q2. Display Rows (4 Marks)
Write R code to:
● Display first 2 rows using head()
● Display last 3 rows using tail()
Q3. Structure and Summary (4 Marks)
Write an R program to:
● Display structure using str()
● Display summary using summary()

Ans :-

```
# -------------------------------
# Practical 3 : dplyr + tibble
# -------------------------------

# Q1. Install and Load dplyr
install.packages("dplyr")
library(dplyr)

# Create dataset
df <- data.frame(
  Name = c("Sujal", "Amit", "Sujal", "Riya", "Amit"),
  Age = c(20, 21, 20, 22, 21)
)

# Display dataset using glimpse()
glimpse(df)
# Rows: 5
# Columns: 2
# $ Name <chr> "Sujal", "Amit", "Sujal", "Riya", "Amit"
# $ Age  <dbl> 20, 21, 20, 22, 21


# Q2. Unique Values and Frequency Table
unique(df$Name)
# [1] "Sujal" "Amit" "Riya"

table(df$Age)
# 20 21 22
#  2  2  1


# Q3. Tibble Conversion
install.packages("tibble")
library(tibble)

tb <- as_tibble(df)
print(tb)
# # A tibble: 5 × 2
#   Name    Age
#   <chr> <dbl>
# 1 Sujal    20
# 2 Amit     21
# 3 Sujal    20
# 4 Riya     22
# 5 Amit     21


# -------------------------------
```

```
# Employee Dataset
# ------------------------------
emp <- data.frame(
  Employee_ID = c(101, 102, 103, 104, 105),
  Employee_Name = c("Sujal", "Amit", "Riya", "Neha", "Raj"),
  Department = c("HR", "IT", "Sales", "Finance", "IT"),
  Salary = c(25000, 30000, 28000, 35000, 32000)
)

# Display dataset
print(emp)
#   Employee_ID Employee_Name Department Salary
# 1         101         Sujal         HR  25000
# 2         102          Amit         IT  30000
# 3         103          Riya      Sales  28000
# 4         104          Neha    Finance  35000
# 5         105           Raj         IT  32000

View(emp)

# Display first 2 rows
head(emp, 2)
#   Employee_ID Employee_Name Department Salary
# 1         101         Sujal         HR  25000
# 2         102          Amit         IT  30000

# Display last 3 rows
tail(emp, 3)
#   Employee_ID Employee_Name Department Salary
# 3         103          Riya      Sales  28000
# 4         104          Neha    Finance  35000
# 5         105           Raj         IT  32000


# Q3. Structure and Summary
str(emp)
# 'data.frame': 5 obs. of  4 variables:
#  $ Employee_ID  : num 101 102 103 104 105
#  $ Employee_Name: chr "Sujal" "Amit" "Riya" "Neha" ...
#  $ Department   : chr "HR" "IT" "Sales" "Finance" ...
#  $ Salary       : num 25000 30000 28000 35000 32000

summary(emp)
#  Employee_ID   Employee_Name      Department           Salary
# Min.   :101   Length:5           Length:5           Min.   :25000
# 1st Qu.:102   Class :character   Class :character   1st Qu.:28000
# Median :103   Mode  :character   Mode  :character   Median :30000
# Mean   :103                                         Mean   :30000
```

```
# 3rd Qu.:104                          3rd Qu.:32000
# Max.   :105                         Max.   :35000
```

Q3.(P4 – Filtering Operations)
Q1. Basic Filtering using subset() (5 Marks)
Create a dataset in R with the following fields:
Product_ID
Product_Name
Price
Category
Display products whose Price is greater than 500 using subset().
Q2. AND Condition Filtering (4 Marks)
Write R code to:
Display Electronics products having Price less than 1000
Use AND (&) condition
Q3. Filter using %in% Operator (4 Marks)
Write R code to:
Display products whose names are Laptop, Mobile, and Tablet
Use %in% operator
Q1. Filter using filter() (5 Marks)
Write R code to:
Filter employees whose Salary is between 20000 and 50000
Use filter() function
Q2. OR Condition Filtering (4 Marks)
Write R code to:
Display employees where Department is HR OR Salary is greater than 60000
Use OR (|) condition
Q3. Average Salary Filtering (4 Marks)
Write R code to:
Calculate average salary
Display employees having Salary greater than or equal to average.

Ans :-

```
# Load dplyr
install.packages("dplyr")
library(dplyr)

# -------------------------------
# Q1. Basic Filtering using subset()
# -------------------------------
products <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Laptop", "Mobile", "Tablet", "Mouse", "Keyboard"),
  Price = c(55000, 15000, 12000, 700, 900),
  Category = c("Electronics", "Electronics", "Electronics", "Accessories", "Accessories")
)

subset(products, Price > 500)
#   Product_ID Product_Name Price     Category
# 1          1       Laptop 55000  Electronics
# 2          2       Mobile 15000  Electronics
```

```
# 3          3       Tablet 12000  Electronics
# 4          4        Mouse   700  Accessories
# 5          5     Keyboard   900  Accessories


# -------------------------------
# Q2. AND Condition Filtering
# -------------------------------
subset(products, Category == "Electronics" & Price < 1000)
# [1] Product_ID   Product_Name Price        Category
# <0 rows> (or 0-length row.names)


# -------------------------------
# Q3. Filter using %in% Operator
# -------------------------------
subset(products, Product_Name %in% c("Laptop", "Mobile", "Tablet"))
#   Product_ID Product_Name Price    Category
# 1          1       Laptop 55000 Electronics
# 2          2       Mobile 15000 Electronics
# 3          3       Tablet 12000 Electronics


# -------------------------------
# Q4. Filter using filter()
# -------------------------------
employees <- data.frame(
  Employee_ID = c(101, 102, 103, 104, 105),
  Employee_Name = c("Sujal", "Amit", "Riya", "Neha", "Raj"),
  Department = c("HR", "IT", "HR", "Finance", "IT"),
  Salary = c(25000, 70000, 40000, 55000, 30000)
)

filter(employees, Salary >= 20000 & Salary <= 50000)
#   Employee_ID Employee_Name Department Salary
# 1         101         Sujal         HR  25000
# 2         103          Riya         HR  40000
# 3         105           Raj         IT  30000


# -------------------------------
# Q5. OR Condition Filtering
# -------------------------------
subset(employees, Department == "HR" | Salary > 60000)
#   Employee_ID Employee_Name Department Salary
# 1         101         Sujal         HR  25000
# 2         102          Amit         IT  70000
# 3         103          Riya         HR  40000
```

```
# -------------------------------
# Q6. Average Salary Filtering
# -------------------------------
avg_salary <- mean(employees$Salary)
avg_salary
# [1] 44000

filter(employees, Salary >= avg_salary)
#   Employee_ID Employee_Name Department Salary
# 1         102         Amit         IT  70000
# 2         104         Neha    Finance  55000
```

Q4.(Based on P5 – Sorting Operations)13 Marks
Q1. Sort Prices using sort() (5 Marks)
Create a dataset in R with fields:
Product_ID
Product_Name
Price
Quantity
Sort the Price in:
Ascending order
Descending order using sort()
Q2. Sort Quantity (4 Marks)
Write R code to:
Sort Quantity in ascending order
Sort Quantity in descending order
Q3. Sort Product Names (4 Marks)
Write R code to:
Sort Product Names alphabetically
Sort Product Names in reverse alphabetical order
Give their output as well

Ans :-

```
# Create Product Dataset
products <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Laptop", "Mobile", "Tablet", "Mouse", "Keyboard"),
  Price = c(55000, 15000, 12000, 700, 900),
  Quantity = c(10, 25, 15, 50, 30)
)

# -------------------------------
```

```
# Q1. Sort Prices using sort()
# -------------------------------
sort(products$Price)
# [1]   700   900 12000 15000 55000

sort(products$Price, decreasing = TRUE)
# [1] 55000 15000 12000   900   700


# -------------------------------
# Q2. Sort Quantity
# -------------------------------
sort(products$Quantity)
# [1] 10 15 25 30 50

sort(products$Quantity, decreasing = TRUE)
# [1] 50 30 25 15 10


# -------------------------------
# Q3. Sort Product Names
# -------------------------------
sort(products$Product_Name)
# [1] "Keyboard" "Laptop" "Mobile" "Mouse" "Tablet"

sort(products$Product_Name, decreasing = TRUE)
# [1] "Tablet" "Mouse" "Mobile" "Laptop" "Keyboard"
```

Q5.(Based on P7 – Selection & Dropping Columns)
Q1. Basic Selection using select() (5 Marks)
Create a dataset named product_data with the following fields:
Product_ID
Product_Name
Category
Price
Stock
Write R code to:
a) Select the columns Product_ID, Product_Name, and Price using select()
b) Display the result
Q2. Column Selection by Position/Pattern (4 Marks)
Write R code to:
a) Select the first three columns using position
b) Select columns starting with the letter P
Q3. Dropping Columns (4 Marks)

Write R code to:
a) Drop the Stock column
b) Drop both Stock and Category columns

Ans :-

```
# Install and load dplyr
install.packages("dplyr")
library(dplyr)

# Create dataset
product_data <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Laptop", "Mobile", "Tablet", "Mouse", "Keyboard"),
  Category = c("Electronics", "Electronics", "Electronics", "Accessories", "Accessories"),
  Price = c(55000, 15000, 12000, 700, 900),
  Stock = c(10, 25, 15, 50, 30)
)

# -------------------------------
# Q1. Basic Selection using select()
# -------------------------------
select(product_data, Product_ID, Product_Name, Price)
#   Product_ID Product_Name Price
# 1          1       Laptop 55000
# 2          2       Mobile 15000
# 3          3       Tablet 12000
# 4          4        Mouse   700
# 5          5     Keyboard   900


# -------------------------------
# Q2. Column Selection by Position/Pattern
# -------------------------------
# First three columns
select(product_data, 1:3)
#   Product_ID Product_Name     Category
# 1          1       Laptop  Electronics
# 2          2       Mobile  Electronics
# 3          3       Tablet  Electronics
# 4          4        Mouse  Accessories
# 5          5     Keyboard  Accessories

# Columns starting with P
select(product_data, starts_with("P"))
#   Product_ID Product_Name Price
# 1          1       Laptop 55000
# 2          2       Mobile 15000
# 3          3       Tablet 12000
```

```
# 4          4       Mouse   700
# 5          5    Keyboard   900


# -------------------------------
# Q3. Dropping Columns
# -------------------------------
# Drop Stock column
select(product_data, -Stock)
#   Product_ID Product_Name     Category Price
# 1          1       Laptop  Electronics 55000
# 2          2       Mobile  Electronics 15000
# 3          3       Tablet  Electronics 12000
# 4          4       Mouse  Accessories   700
# 5          5    Keyboard  Accessories   900

# Drop Stock and Category columns
select(product_data, -Stock, -Category)
#   Product_ID Product_Name Price
# 1          1       Laptop 55000
# 2          2       Mobile 15000
# 3          3       Tablet 12000
# 4          4       Mouse   700
# 5          5    Keyboard   900
```

Q6.(Based on P9 – String Operations)
Q1. String Creation and Extraction (5 Marks)
Write R code to:
a) Install and load the stringr package
b) Create a string "Technology"
c) Extract the first character and last character using str_sub()
Q2. Substring Extraction (4 Marks)
Write R code to:
a) Extract a substring from "ComputerScience" (position 5 to 14)
b) Display the result
Q3. String Splitting (4 Marks)
Write R code to:
a) Split the sentence "Python is simple and useful" into words
b) Split "Red,Blue,Green,Yellow" using comma

Ans :-

```
# Install and load stringr
install.packages("stringr")
library(stringr)
```

```
# -------------------------------
# Q1. String Creation and Extraction
# -------------------------------
text1 <- "Technology"

# First character
str_sub(text1, 1, 1)
# [1] "T"

# Last character
str_sub(text1, -1, -1)
# [1] "y"


# -------------------------------
# Q2. Substring Extraction
# -------------------------------
text2 <- "ComputerScience"

str_sub(text2, 5, 14)
# [1] "uterScienc"


# -------------------------------
# Q3. String Splitting
# -------------------------------
sentence <- "Python is simple and useful"
str_split(sentence, " ")
# [[1]]
# [1] "Python" "is" "simple" "and" "useful"

colors <- "Red,Blue,Green,Yellow"
str_split(colors, ",")
# [[1]]
# [1] "Red"    "Blue"   "Green"  "Yellow"
```

Q7(Based on practical 10)
Q1. Create a dataset (3 Marks)
Create a data frame with:
- ID
- Name
- Marks

● Attendance
Q2. Perform Calculations (5 Marks)
1. Create Percentage
2. Create Bonus_Marks
○ If Attendance > 90 → 5 marks
○ Otherwise → 0
3. Create Final_Marks (Marks + Bonus_Marks)
Q3. Advanced Variables (5 Marks)
1. Create Performance Category using cut()
○ 0–60 → Average
○ 60–75 → Good
○ 75–90 → Very Good
○ 90–100 → Excellent
2. Create Eligible_Scholarship
○ TRUE if Final_Marks ≥ 85
3. Create Rank based on Final_Marks

Ans :-

```
# Install and load dplyr
install.packages("dplyr")
library(dplyr)

# -------------------------------
# Q1. Create Dataset
# -------------------------------
student_data <- data.frame(
  ID = c(1, 2, 3, 4, 5),
  Name = c("Sujal", "Amit", "Riya", "Neha", "Raj"),
  Marks = c(78, 92, 67, 88, 55),
  Attendance = c(95, 85, 91, 98, 80)
)

# -------------------------------
# Q2. Perform Calculations
# -------------------------------
student_data <- student_data %>%
  mutate(
    Percentage = Marks,
    Bonus_Marks = ifelse(Attendance > 90, 5, 0),
    Final_Marks = Marks + Bonus_Marks
  )

# -------------------------------
# Q3. Advanced Variables
# -------------------------------
student_data <- student_data %>%
  mutate(
    Performance = cut(
```

```
      Final_Marks,
      breaks = c(0, 60, 75, 90, 100),
      labels = c("Average", "Good", "Very Good", "Excellent"),
      include.lowest = TRUE
    ),
    Eligible_Scholarship = Final_Marks >= 85,
    Rank = rank(-Final_Marks)
  )

# Display final dataset
print(student_data)

# Output:
#   ID  Name Marks Attendance Percentage Bonus_Marks Final_Marks Performance
Eligible_Scholarship Rank
# 1  1 Sujal    78         95         78           5          83   Very Good              FALSE    3
# 2  2  Amit    92         85         92           0          92   Excellent              TRUE    1
# 3  3  Riya    67         91         67           5          72        Good              FALSE    4
# 4  4  Neha    88         98         88           5          93   Excellent              TRUE    0
# 5  5   Raj    55         80         55           0          55     Average              FALSE    5
```

Q8(Based on practical 11)
Q1. Create Product Dataset (3 Marks)
Create a wide format dataset with the following fields:
● Product Name
● Sales for 3 months: January, February, March
Write an R program to create the dataset.
Q2. Reshape Data (6 Marks)
Perform the following operations:
● Convert the dataset from wide format to long format using pivot_longer()
● Create the columns:
○ Month
○ Sales
Display the long format dataset.
Q3. Convert Back (4 Marks)
Perform the following operation:
● Convert the long format dataset back to wide format using pivot_wider()
● Print the final dataset.

Ans :-

```
# Install and load tidyr
install.packages("tidyr")
library(tidyr)
```

```
# -------------------------------
# Q1. Create Wide Format Dataset
# -------------------------------
product_sales <- data.frame(
  Product_Name = c("Laptop", "Mobile", "Tablet"),
  January = c(50000, 30000, 20000),
  February = c(55000, 32000, 22000),
  March = c(60000, 35000, 25000)
)

print(product_sales)
#   Product_Name January February March
# 1       Laptop   50000    55000 60000
# 2       Mobile   30000    32000 35000
# 3       Tablet   20000    22000 25000


# -------------------------------
# Q2. Convert Wide to Long
# -------------------------------
long_data <- pivot_longer(
  product_sales,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)

print(long_data)
#   Product_Name    Month    Sales
# 1       Laptop   January   50000
# 2       Laptop  February   55000
# 3       Laptop     March   60000
# 4       Mobile   January   30000
# 5       Mobile  February   32000
# 6       Mobile     March   35000
# 7       Tablet   January   20000
# 8       Tablet  February   22000
# 9       Tablet     March   25000


# -------------------------------
# Q3. Convert Long to Wide
# -------------------------------
wide_data <- pivot_wider(
  long_data,
  names_from = Month,
  values_from = Sales
```

```
)

print(wide_data)
#   Product_Name January February March
# 1      Laptop   50000    55000 60000
# 2      Mobile   30000    32000 35000
# 3      Tablet   20000    22000 25000
```

Q9(Based on practical 12)
Q1. Combine Employee Datasets (13 Marks)
(a) Create Two Data Frames (3 Marks)
Create two data frames employee_A and employee_B with the following variables:
● Emp_ID
● Name
● Salary
(Use at least 3 records in first dataset and 2 records in second dataset)
(b) Combine Datasets using rbind() (6 Marks)
Write R code to combine employee_A and employee_B using the rbind() function and display the final dataset.
(c) Explain rbind() (4 Marks)
Explain the concept of rbind() in R. Also mention the conditions required for combining datasets using rbind().

Ans :-

```
# -------------------------------
# Q1(a) Create Two Data Frames
# -------------------------------
employee_A <- data.frame(
  Emp_ID = c(101, 102, 103),
  Name = c("Sujal", "Amit", "Riya"),
  Salary = c(25000, 30000, 28000)
)

employee_B <- data.frame(
  Emp_ID = c(104, 105),
  Name = c("Neha", "Raj"),
  Salary = c(35000, 32000)
)

print(employee_A)
#   Emp_ID  Name Salary
# 1    101 Sujal  25000
# 2    102  Amit  30000
# 3    103  Riya  28000
```

```
print(employee_B)
#   Emp_ID Name Salary
# 1    104 Neha  35000
# 2    105  Raj  32000


# -------------------------------
# Q1(b) Combine using rbind()
# -------------------------------
final_employee <- rbind(employee_A, employee_B)

print(final_employee)
#   Emp_ID  Name Salary
# 1    101 Sujal  25000
# 2    102  Amit  30000
# 3    103  Riya  28000
# 4    104 Neha  35000
# 5    105  Raj  32000


# -------------------------------
# Q1(c) Explain rbind()
# -------------------------------
# rbind() means row bind.
# It is used to combine two or more data frames by adding rows one below another.
#
# Conditions required:
# 1. Both data frames must have same number of columns
# 2. Column names should be same
# 3. Data types of corresponding columns should be same
```

Q10(Based on practical 13)
Q1. Create Dataset (3 Marks)
Create a dataset in R with duplicate records containing:
● Product_ID
● Product_Name
● Price
Q2. Identify Duplicate Rows (2 Marks)
Write an R command to identify duplicate rows in the dataset.
Q3. Display Duplicate Records (2 Marks)
Write an R command to display only the duplicate records.
Q4. Remove Duplicate Records (3 Marks)
Write an R command to remove duplicate records using an appropriate function.
Q5. Remove Duplicates based on Product_ID (2 Marks)
Write an R command to remove duplicates based only on Product_ID while keeping all

other columns.
Q6. Count Duplicate Records (1 Mark)
Write an R command to count the number of duplicate records in the dataset.

Ans :-

```
# -------------------------------
# Q1. Create Dataset with Duplicates
# -------------------------------
products <- data.frame(
  Product_ID = c(1, 2, 2, 3, 4, 4),
  Product_Name = c("Laptop", "Mobile", "Mobile", "Tablet", "Mouse", "Mouse"),
  Price = c(55000, 15000, 15000, 12000, 700, 700)
)

print(products)
#   Product_ID Product_Name Price
# 1          1       Laptop 55000
# 2          2       Mobile 15000
# 3          2       Mobile 15000
# 4          3       Tablet 12000
# 5          4        Mouse   700
# 6          4        Mouse   700


# -------------------------------
# Q2. Identify Duplicate Rows
# -------------------------------
duplicated(products)
# [1] FALSE FALSE  TRUE FALSE FALSE  TRUE


# -------------------------------
# Q3. Display Only Duplicate Records
# -------------------------------
products[duplicated(products), ]
#   Product_ID Product_Name Price
# 3          2       Mobile 15000
# 6          4        Mouse   700


# -------------------------------
# Q4. Remove Duplicate Records
# -------------------------------
unique(products)
#   Product_ID Product_Name Price
# 1          1       Laptop 55000
# 2          2       Mobile 15000
# 4          3       Tablet 12000
```

```
# 5          4        Mouse   700


# --------------------------------
# Q5. Remove Duplicates based on Product_ID
# --------------------------------
products[!duplicated(products$Product_ID), ]
#   Product_ID Product_Name Price
# 1          1       Laptop 55000
# 2          2       Mobile 15000
# 4          3       Tablet 12000
# 5          4        Mouse   700


# --------------------------------
# Q6. Count Duplicate Records
# --------------------------------
sum(duplicated(products))
# [1] 2
```

Q11(Based on practical 14)
Q1. (3 Marks)
Create a dataset with a Date column and display it.
Q2. (2 Marks)
Write a command to extract the week number of the year from the dates.
Q3. (2 Marks)
Extract the month name from the date column.
Q4. (2 Marks)
Extract the day and weekday from the dataset.
Q5. (2 Marks)
Add new columns (Year, Month, Day, Weekday) to the dataset.
Q6. (1 Mark)
What function is used to extract the year in lubridate?
Q7. (1 Mark)
Why is date conversion important in data analysis?

Ans :-

```
# Install and load lubridate
install.packages("lubridate")
library(lubridate)

# --------------------------------
# Q1. Create Dataset with Date
# --------------------------------
date_data <- data.frame(
```

```
  Date = as.Date(c("2026-01-15", "2026-03-22", "2026-07-10"))
)

print(date_data)
#         Date
# 1 2026-01-15
# 2 2026-03-22
# 3 2026-07-10


# -------------------------------
# Q2. Extract Week Number
# -------------------------------
week(date_data$Date)
# [1] 3 12 28


# -------------------------------
# Q3. Extract Month Name
# -------------------------------
month(date_data$Date, label = TRUE)
# [1] Jan Mar Jul


# -------------------------------
# Q4. Extract Day and Weekday
# -------------------------------
day(date_data$Date)
# [1] 15 22 10

wday(date_data$Date, label = TRUE)
# [1] Thu Sun Fri


# -------------------------------
# Q5. Add New Columns
# -------------------------------
date_data$Year <- year(date_data$Date)
date_data$Month <- month(date_data$Date, label = TRUE)
date_data$Day <- day(date_data$Date)
date_data$Weekday <- wday(date_data$Date, label = TRUE)

print(date_data)
#         Date Year Month Day Weekday
# 1 2026-01-15 2026   Jan  15     Thu
# 2 2026-03-22 2026   Mar  22     Sun
# 3 2026-07-10 2026   Jul  10     Fri
```

```
# --------------------------------
# Q6. Function for Year
# --------------------------------
# year()


# --------------------------------
# Q7. Importance of Date Conversion
# --------------------------------
# Date conversion is important because it helps in:
# 1. Correct sorting of dates
# 2. Extracting year, month, day
# 3. Time-based analysis and reports
```

Module 2

Q12(Based on practical 1 OF MODULE 2)[12 MARKS] Q1. Create Data Frame (3 Marks)
Write an R program to create a data frame named student_data containing marks of two subjects.
Q2. Descriptive Statistics using summary() (3 Marks)
Write an R program to generate the summary of the data frame using the summary() function. Q3. Descriptive Statistics using describe() (3 Marks)
Write an R program to install and load the psych package and apply the describe() function on the dataset.
Q4. Minimum, Maximum and Range (3 Marks)
Write an R program to find the minimum, maximum, and range of Maths marks from the dataset.

Ans :-

```
# --------------------------------
# Q1. Create Data Frame
# --------------------------------
student_data <- data.frame(
  Student_ID = c(1, 2, 3, 4, 5),
  Maths = c(78, 85, 90, 88, 76),
  Science = c(82, 79, 91, 87, 80)
)

print(student_data)
#   Student_ID Maths Science
# 1          1    78      82
# 2          2    85      79
# 3          3    90      91
# 4          4    88      87
# 5          5    76      80


# --------------------------------
# Q2. Descriptive Statistics using summary()
# --------------------------------
summary(student_data)
#   Student_ID       Maths          Science
# Min.   :1    Min.   :76.0   Min.   :79.0
# 1st Qu.:2    1st Qu.:78.0   1st Qu.:80.0
# Median :3    Median :85.0   Median :82.0
# Mean   :3    Mean   :83.4   Mean   :83.8
# 3rd Qu.:4    3rd Qu.:88.0   3rd Qu.:87.0
# Max.   :5    Max.   :90.0   Max.   :91.0


# --------------------------------
```

```
# Q3. Descriptive Statistics using describe()
# -------------------------------
install.packages("psych")
library(psych)

describe(student_data)
# Output includes:
# variables, n, mean, sd, median, min, max, range, skew, kurtosis


# -------------------------------
# Q4. Minimum, Maximum and Range
# -------------------------------
min(student_data$Maths)
# [1] 76

max(student_data$Maths)
# [1] 90

range(student_data$Maths)
# [1] 76 90
```

Q13(Based on practical 2 OF MODULE 2)[12 MARKS] Q1. Create a sample dataset named students with variables Gender, Course, and Grade. Display the dataset. (3 Marks)
Q2. Write an R program to generate a one-way frequency table for the variable Gender using table() function. (3 Marks)
Q3. Write an R program to generate a frequency table for the variable Course using table() function. (3 Marks)
Q4. Write an R program to count the frequency of Gender using count() function from dplyr package. (3 Marks)

Ans :-

```
# Install and load dplyr
install.packages("dplyr")
library(dplyr)

# -------------------------------
# Q1. Create Sample Dataset
# -------------------------------
students <- data.frame(
  Gender = c("Male", "Female", "Male", "Female", "Male"),
  Course = c("BCA", "BBA", "BCA", "BCom", "BBA"),
  Grade = c("A", "B", "A", "C", "B")
```

```
)

print(students)
#   Gender Course Grade
# 1   Male    BCA     A
# 2 Female    BBA     B
# 3   Male    BCA     A
# 4 Female   BCom     C
# 5   Male    BBA     B


# --------------------------------
# Q2. Frequency Table for Gender
# --------------------------------
table(students$Gender)
# Female   Male
#      2      3


# --------------------------------
# Q3. Frequency Table for Course
# --------------------------------
table(students$Course)
# BBA BCA BCom
#   2   2    1


# --------------------------------
# Q4. Count Gender using dplyr
# --------------------------------
count(students, Gender)
#   Gender n
# 1 Female 2
# 2   Male 3
```